# Tzu-Chun Hsu

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# Work Experience

## Camera ISP Firmware Engineer

Realtek Semiconductor Corp.

Hsinchu, Taiwan 02 2025 – present

1. **ISP Development & Algorithm Optimization.**
2. **Software Architecture & Development.**
3. **Test Automation & Verification.**

# Work Experience

## Camera ISP Firmware Engineer

Realtek Semiconductor Corp.

Hsinchu, Taiwan 02 2025 – present

1. **ISP Development & Algorithm Optimization:**
   - Developed high-resolution image stitching solution for smart glasses.
   - Aligned ISP pipeline for snapshot and streaming capture to ensure consistent output.
   - Migrated ISP engine for smart glasses from Linux to FreeRTOS.
   - Implemented ISP pipeline features for smart glasses with V4L2 and interrupt controls and optimized SDK flow.
2. **Software Architecture & Development:**
3. **Test Automation & Verification:**

# Work Experience

## Camera ISP Firmware Engineer

Realtek Semiconductor Corp.

Hsinchu, Taiwan 02 2025 – present

1. **ISP Development & Algorithm Optimization:**
2. **Software Architecture & Development:**
   - Refactored core CModel architecture and consolidated sub-applications.
   - Redesigned more detailed and readable image format table.
   - Implemented thread and cross-process synchronization for Windows.
3. **Test Automation & Verification:**

# Work Experience

## Camera ISP Firmware Engineer

Realtek Semiconductor Corp.

Hsinchu, Taiwan 02 2025 – present

1. **ISP Development & Algorithm Optimization:**
2. **Software Architecture & Development:**
3. **Test Automation & Verification:**
   - Designed and optimized automated verification systems for ISP/IPU (FPGA/ASIC) hardware platforms.
   - Migrated communication protocol from JRPC to protobuf for the Linux-to-FreeRTOS migration.

# Education

## Zhejiang University

Bachelor of Engineering in Computer Science and Technology
Hangzhou, China 09 2016 – 07 2020

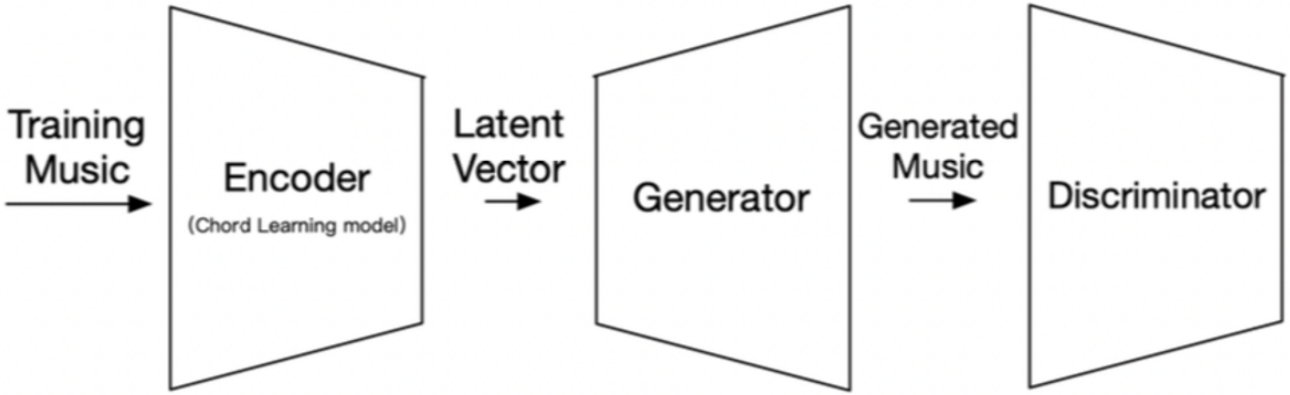
## National Yang Ming Chiao Tung University

Master of Science in Computer Science and Engineering
Hsinchu, Taiwan 09 2022 – 02 2025

# Projects

## 1. Chord learning and adversarial framework for symbolic music generation

- Proposed a chord learning framework for multi-track symbolic music generation based on VAE and GAN.
- Introduced WGAN-GP to solve mode collapse and vanishing gradient problem.



# Projects

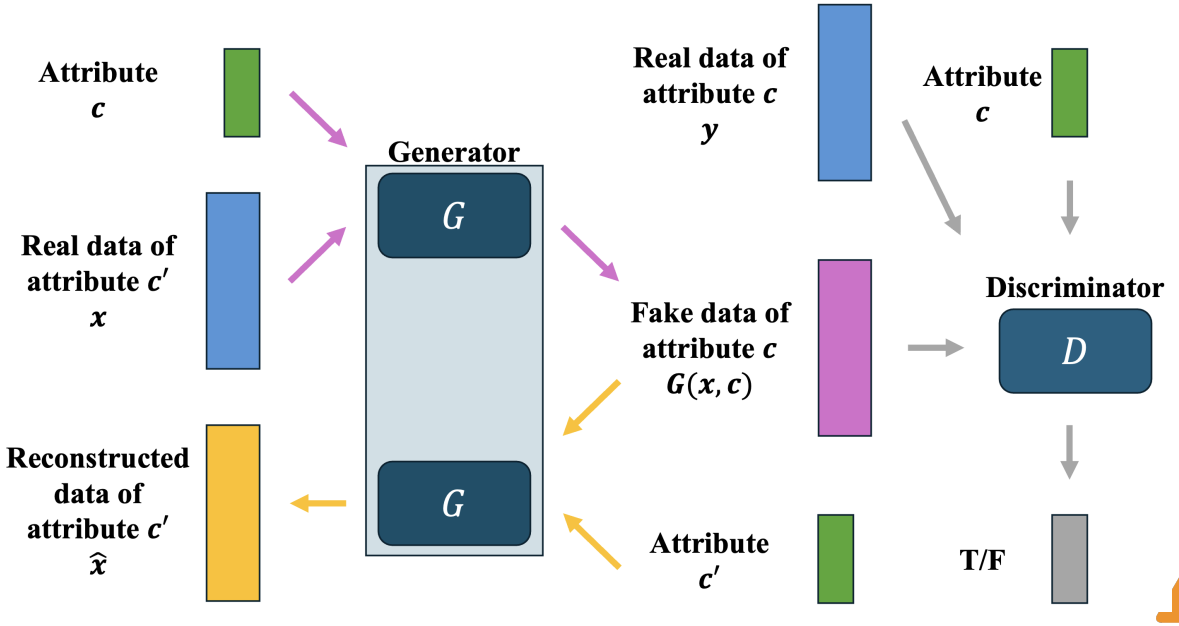
## 1. Chord learning and adversarial framework for symbolic music generation

- ✓ CCCC-Mobile Application Innovation Contest
  *First Prize and Most Innovative Award*
- ✓ The 4th Internet+ Innovation and Entrepreneurship Competition
  *Gold Award*

# Projects

## 2. (Bachelor's Thesis) Voice Conversion Based on Generative Adversarial Networks

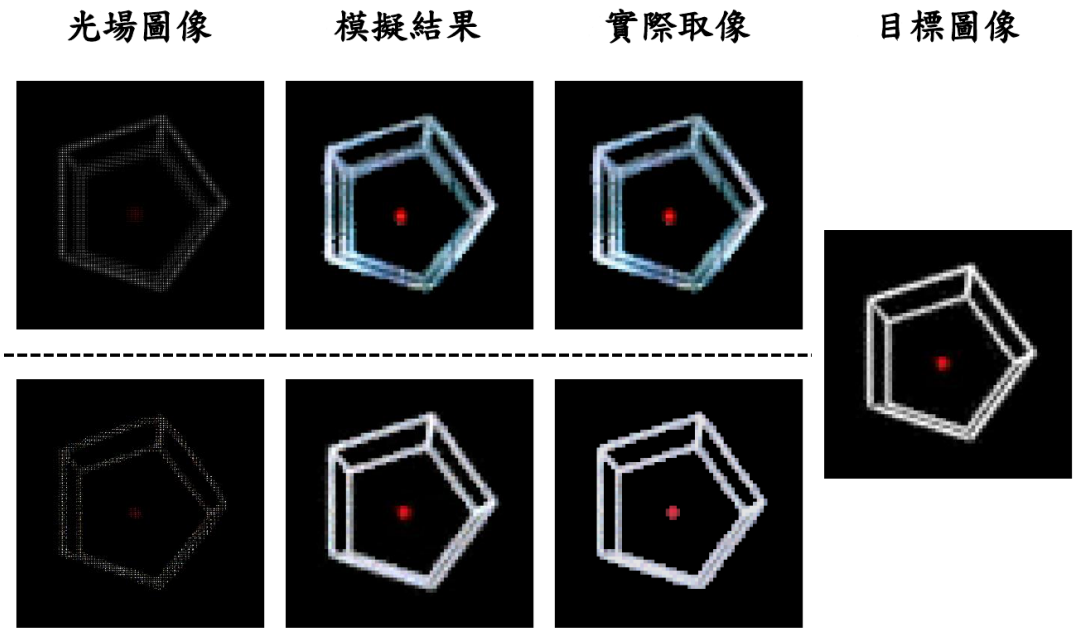
- Improved StarGAN-VC2 based on multi-speaker non-parallel corpus.
- Introduced WGAN-div, AdaIN layer, and neural vocoder, WaveGlow, for better results.



# Projects

## 3. (Master's Thesis) Floating image quality compensation algorithm technology

- Proposed a two-stage method to enhanced image quality of floating images: (1) simulating imaging process by solving QP problem derived from optical principles, and (2) refining light field images iteratively based on the simulation.



# Skills

## Languages

Python, C, C++, Java, JavaScript

## Technologies/Frameworks

Flask, PyTorch, Spring Boot, Vue.js

# Thanks for your attention

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